

**Smart Home IoT: Sensor Integration
and Smart Control Based on ESP32**



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Objective

This project aimed to design and develop a smart home system using the ESP32 microcontroller integrated with sensors and a mobile application. It was intended to improve home security, user comfort, and energy efficiency through real-time monitoring and remote control.

Tools and Materials

- **Tools:**

- Laptop / Computer
- Arduino IDE
- Smartphone (Blynk application)
- USB cable

- **Materials:**

- ESP32 microcontroller
- PIR sensor
- DHT11 sensor
- MQ-2 gas sensor
- RFID module and card
- Rain sensor
- Servo motors
- Buzzer
- LED lights
- Jumper wires
- Breadboard

Procedure

1. The system was planned and designed.
2. The tools and materials were prepared.
3. The hardware components were assembled.
4. The program was written and uploaded using Arduino IDE.
5. The system was integrated with the Blynk application.
6. The system was tested and evaluated to ensure proper functionality and performance.

Result

The system operated successfully in accordance with the planned design. The PIR sensor detected motion effectively, while the RFID module controlled secure door access. The DHT11 sensor monitored temperature and humidity accurately, and the MQ-2 sensor detected gas leakage reliably. In addition, the rain sensor automatically adjusted the clothesline based on weather conditions. All components were integrated with the ESP32 and connected to the Blynk application, allowing real-time remote monitoring and control. The system demonstrated stable and reliable performance during testing.



Conclusion

The project was successfully completed, and all system components functioned as expected. The integration of ESP32 with various sensors and the Blynk application enabled effective real-time monitoring and remote control. The system improved home security, comfort, and energy efficiency. In addition, this project provided practical experience and improved understanding of IoT technology, electronics, and programming in a smart home system.